MILITARY SPECIFICATION

CHARGER, BATTERY, PP-7286/U

This specification is approved for use by Electronic Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of battery charger designated PP-7286/U and Case, Battery Charger, CY7670/U. The Charger is capable of individually or simultaneously charging five secondary batteries at constant currents from 15 to 700 millamperes and at terminal voltages from 0 to 41 volts when powered by a nominal 115/230 Vac, 50/60/400 Hz source. (See 6.1, 6.5)

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation-Packaging, Methods of
MIL-P-11268	Parts, Materials, and Processes used in Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-E-55585	Electronic Equipment and Parts, Packaging of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Electronics Research and Development Command, ATTN: DELET-R-S, Fort Monmouth, New Jersey 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6130

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurements of
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-810	Environmental Test Methods
MIL-STD-726	Packaging Requirement Codes

DRAWINGS

Electronics Command

DL-SM-B-889033	Set Assembly Battery Charger

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3.

3.2 Construction. The equipment shall be constructed in accordance with Drawing DL-SM-B-889033 (see 4.10).

3.3 Parts, materials, and processes; general. In addition, to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein, shall apply (see 4.10).

3.4 System safety.

3.4.1 Personnel hazards. Personnel hazards shall be kept to a minimum through compliance of configuration changes and parts selection with Requirement 1 of MIL-STD-454. Compliance with these requirements will be verified through a visual inspection (see 4.10).

3.4.2 Edge rounding exposed. Exposed edges shall be rounded as specified in the equipment drawings.

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3.4.3 Radioactive materials. Radioactive materials shall not be used (e.g., luminous dials/markings, electron tubes, surge arrestors and lenses).

3.4.4 Output terminals. Output terminal voltage shall not exceed 70 VDC under open circuit conditions (see 4.7).

3.5 Marking. Marking shall conform to MIL-M-13231. Front panel marking shall be Group I as described in that specification.

3.5.1 Visibility. Wherever practicable, parts and assemblies shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.5.2 Serial number. Each battery charger shall be serially numbered.

3.6 Finish. Equipment shall be given protective finish in accordance with MIL-F-14072. This includes finish of hardware such as hinges, screws, etc., and necessary touch up after mounting. The final paint film on Type I surface shall be forest green enamel, film designation per MIL-F-14072.

3.7 Electrical characteristics. (See 4.7)

3.7.1 Input. The input voltage shall be 115/230 VAC ± 10%, 50/60/400 Hz, ± 5% single phase.

3.7.2 Output.

3.7.2.1 Output, current settability. The output consists of five constant current output channels. Each channel is individually current regulated and capable of providing 15 to 700 mA. Each channel shall be continuously adjustable throughout the cited range.

3.7.2.2 Output current regulation and compliance voltage. Each channel shall have an output current regulated to +5 -10% relative to the current at nominal line voltage and short circuit load and shall have sufficient compliance voltage to maintain the output current for all channels under the combined variables of input voltage, frequency, and load conditions covering short circuit up to the highest range of battery back e.m.f. of 41 VDC.

3.7.2.3 Output current and voltage measurement. The extended scale panel meter and switching shall permit measurement of the output voltage and current of each charging channel. Current measurement scales shall be 0-50, 0-200 and 0-700mA corresponding to the adjustment range switch setting for the channel being measured. Channel output voltage measurement scale be 0-50 volts. Actuation of a momentary switch shall be required to connect the voltmeter to the charging channels or the test load and the voltmeter to the memory battery. Full scale meter range shall be 10 Volts for memory battery test.

3.7.3 Isolation. The output of all five channels shall be electrically isolated from the input and ground.

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3.7.4 Digital timer.

3.7.4.1 Controls/display. The timer shall be capable of being set from 00.0 to 19.9 in one-tenth hour increments. Additionally the timer shall initiate charging current on all channels and shall terminate charging current when the timer display decreases below .1, at which point the display will blank.

3.7.4.2 Memory battery. Means shall be provided for checking the memory battery to assure a minimum terminal voltage of 7.0 VDC.

3.7.4.3 Timer accuracy and memory test. The timer shall be accurate to within 5% of its preset value. In the event of source input power interruption the timer shall resume timing upon return of power.

3.7.5 Battery discharge into charger. The equipment shall inhibit a discharge current in excess of 100 ua from the battery being charged under the conditions of:

a. Loss of power input to the charger.
b. Battery emf in excess of charger DC output voltage.

3.8 Environmental service conditions.

3.8.1 Altitude. (See 4.8.1)

a. The equipment shall be capable of operating in elevations up to 10,000 feet (3000M) above sea level.

b. The equipment shall be capable of withstanding non-operating modes in elevations up to 40,000 feet (12,000 meters) above sea level.

3.8.2 Temperature. (see 4.8.2)

a. Operating. The equipment shall operate, without degradation in specified performance, at ambient temperatures in the range of -25°F (-32°C) to +145°F (63°C).

b. Non-operating. The equipment shall be operable, without degradation in specified performance, after exposure to ambient temperatures in the range of -70°F (-57°C) to +160°F (71°C).

3.8.3 Humidity. (See 4.8.3) The equipment shall be operable without degradation in specified performance, and shall sustain no physical damage, during and after prolonged exposure to extreme high humidity levels as encountered in tropical areas.

3.8.4 Dust. (See 4.8.4) The equipment, in both operating and non-operating condition, shall withstand exposure to dust particles with wind speed of 17 knots (31.5 km per hour).

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3.8.5 Fungus. (See 4.8.5) The equipment, in the assembled and ready for delivery condition shall provide no nutrients in material, coating, or contaminant form nor support fungal growth when tested in accordance with 4.8.5. Only inherently fungus resistant grades of materials per Requirement 4, MIL-STD-454, shall be used. Prior to delivery, all fungus nutrient contaminants or residues shall be removed.

3.8.6 Vibration and shock. (See 4.8.6) The charger in its transit case shall withstand vibration and shock induced during field transport by military vehicles and by rough handling. The equipment shall withstand shock encountered during servicing.

3.8.7 Salt fog. (See 4.8.7) The equipment shall be resistant to the corrosive effects of salt atmosphere.

3.8.8 Immersion. (See 4.8.8) The equipment in its transit case shall be immersion proof to a covering depth of 3 feet (91 cm) of water for a period of not less than 2 hours.

3.9 Electromagnetic interference reduction. (See 4.9) The equipment shall meet the emission and susceptibility requirements of MIL-STD-461A, Notice 4, listed below:

a. CE02 10KHz to 50 KHz
b. CE04 50KHz to 50 MHz broadband and narrowband
c. CS02 50KHz to 400MHz
d. CS06 Spike, power leads
e. RE02 14KHz to 1 GHz broadband
f. RE02.1 14KHz to 10GHz narrowband
g. RS02 Magentic induction field spike
h. RS03

	Freq Test Range	"E" field	Modulation Characteristics
	10KHz - 1.9 MHz	1 V/M	AM, 50%, 1KHz tone
	2MHz - 29.99MHz	5 V/M	AM, 50%, 1KHz tone
	30MHz - 75.99MHz	10 V/M	FM, 10KHz Dev, 1 KHz tone
	76MHz - 399MHz	10 V/M	AM, 50%, 1KHz tone
RS03.1	400MHz - 1.9 GHz	10 V/M	Pulse, .1microsec, 400PPS
	2GHz - 12.4GHz	5 V/M	Pulse, .1microsec, 400PPS

NOTE: The equipment shall comply with subparagraph 3.7.2.2 when subjected to the above radiated environments.

3.9.1 Bonding and Grounding. (See 4.9.1) The equipment shall be designed and constructed with adequate provisions for grounding the unit. The bonding of subassemblies and components to frames and chassis shall not be impeded by non-conductive protective finishes. Bonding jumpers, when required, shall be of the solid metal type and be as short as possible. The D.C. resistance of bonds and grounds at any single mating surface shall not exceed 2.4 Milliohms.

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3.10 Reliability. (see 4.5.4.5) The equipment shall have a MTBF of 2,000 hrs when used in the following manner:

Ambient temperature +100°F (38°C)
Duty Cycle - continuous
Environment - ground

The reliability shall be test demonstrated in accordance with 4.5.4.

3.11 Burn-in. See 4.6) The equipment shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the burn-in of 4.6. Each equipment on order shall be subjected to a burn-in of a minimum of 48 hours with the last 24 hours failure free.

3.12 Interchangeability. Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment (see 4.11). Individual items shall not be handpicked for fit or performance. Reliance shall not be placed on any unspecified dimensions, rating, characteristic, etc.

3.13 Workmanship. (see 4.10) Workmanship shall be in accordance with requirement 9 of MIL-STD-454.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (does not include packaging) (see 4.3).
b. Inspection covered by subsidiary documents. (see 4.4).
c. Quality conformance inspection.

1. Quality conformance inspection of equipment before packaging. (see 4.5).
2. Quality conformance inspection of packaging. (see 4.12).

4.3 First article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish six (6) first article units of the complete Battery Charger PP-7286/U unless otherwise specified in the contract.

4.3.2 First article inspection. The first article inspection shall consist of inspections specified in Table I, and shall be performed in the order specified therein.

4.3.3 First article data. The first article test plan and test report(s) shall be as required in the contract.

4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

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TABLE I FIRST ARTICLE INSPECTION

Item	Inspection	Reqm't Para.	Test	Order of Test					
				Unit 1	Unit 2	Unit 3	Unit 4	Unit 5	Unit 6
1.	Inspection covered by subsidiary documents	3.3, 3.5, 3.6	4.4	Inspections to be performed on all units					
2.	Burn-in	3.11	4.6	Inspections to be performed on all units					
3.	Group A	See Table II		Inspections to be performed on all units					
4.	Group B	See Table III		Inspections to be performed on all units					
5.	EMI	3.9	4.9			1			
6.	Environmental	3.8	4.8						
	Altitude	3.8.1	4.8.1	1					
	Temperature	3.8.2	4.8.2	2					
	Humidity	3.8.3	4.8.3	3					
	Salt Fog	3.8.7	4.8.7	4					
	Immersion	3.8.8	4.8.8		2,5				
	Vibration	3.8.6	4.8.6		3				
	Shock	3.8.6	4.8.6		4				
	Dust	3.8.4	4.8.4		1				
	Fungus	3.8.5	4.8.5			2			
7.	Group D								
	Reliability	3.10	4.5.4				1	1	1

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Item	Where Required
Parts materials, and processes general	3.3
Marking	3.5
Finish, protective	3.6

4.5 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspection specified in 4.4, 4.5.1 and 4.6 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit on contract shall be inspected for conformance to the inspections specified in Table II. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract, shall be in accordance with MIL-STD-105, using the general inspection levels, and AQL's indicated in Table II.

TABLE II GROUP A INSPECTION

Inspection	Requirement Paragraph	Insp. Para.	AQL Major	AQL Minor
Visual and mechanical	3.2, 3.3, 3.4.1 3.13	4.10	1.5 dphu*	6.5
Electrical			1 dphu for the group	**
Output	3.7.2	4.7		
Digital Timer	3.7.4	4.7		
Input	3.7.1	4.7		

* dphu - defects per hundred units
** All electrical defects are considered major.

4.5.1.1 Order of inspection within group A. Group A inpsection shall be performed in the order specified in Table II.

4.5.2 Group B inspection. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table III and to the special inspection levels of Table I of MIL-STD-105.

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TABLE III GROUP B INSPECTION

Inspection	Req. Para.	Insp. Para.	AQL
Isolation	3.7.3	4.7	4.0%
Battery discharge into Charger	3.7.5	4.7	4.0%
Interchangeability	3.12	4.11	6.5%
Output	3.7.2	4.7	4.0%

4.5.2.1 Group B sampling plans. The Group B sampling plans, for the AQLs listed in Table III shall be as follows:

AQL	Inspection Level
4.0 dphu	S-4
6.5 dphu	S-4

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table IV and shall be performed in the order shown. Samples shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for group C inspection.

4.5.3.1.1 Three (3) samples of the completed equipment shall be selected at random, One (1) sample for each subgroup I through III in Table IV. The samples shall be selected at the start of the contract from the first quality conformance inspection lot. (Subgroup III tests shall not be performed when First Article samples are specified by contract.)

4.5.3.1.2 Thereafter, two (2) samples of the complete equipment shall be selected at random, one sample for each subgroup I and II in Table IV from every 200 units or fraction thereof greater than or equal to one-half (½). If the number of units remaining for the final Group C inspections is less than 100 units, these units shall be included in the selection of samples for preceding Group C inspection. The maximum lot size for the final Group C inpsection shall be limited to 300 units.

4.5.3.2 Group C failure. Actions required relative to Group C failures shall be as specified in the contract.

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4.5.3.3 Reinspection of conforming group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspections may be accepted on the contract or purchase order provided all visible and known damage is repaired and the sample units are resubjected to and pass Group A inspection.

Table IV Group C Inspection

Inspection	Req. Para.	Insp. Para.
Group I		
Altitude	3.8.1	4.8.1
Low temperature	3.8.2	4.8.2
High temperature	3.8.2	4.8.2
Humidity	3.8.3	4.8.3
Group II		
Dust	3.8.4	4.8.4
Immersion*	3.8.8	4.8.8
Vibration	3.8.6	4.8.6
Shock	3.8.6	4.8.6
Salt fog	3.8.7	4.8.7
Group III		
EMI	3.9	4.9
Fungus	3.8.5	4.8.5

*Performed prior to and following vibration and shock testing.

4.5.4 Group D inspection. This inspection shall consist of the test specified in Table V and shall be performed on units from lots which have been subjected to and met Groups A and B inspection. Test conditions, methods, and procedures shall be in accordance with paragraph 4.5.4.6. Individual units which have been subjected to potentially damaging environmental tests will not be selected for realiability testing.

Table V Group D Inspection

Inspection	Req. Para.	Insp. Para.
Reliability	3.10	4.5.4

4.5.4.1 Sampling for group D inspection

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4.5.4.1.1 Sampling for first article reliability qualification. For the purpose of First Article Qualifications, three (3) units shall be tested.

4.5.4.1.2 Sampling for quality conformance (group D) inspection. For the purposes of Quality Conformance inspection, at least three (3) and no more than ten (10) units shall be selected at random from each lot. Unless otherwise specified, for reliability testing purposes, a lot is defined as the greater of one month's production or 200 units. If the number of units remaining for the final Group D inspection is less than 100 units, these units shall be included in the selection of samples for preceding Group D inspection. The maximum lot size for the final Group D inspection shall be 300 units.

4.5.4.1.3 Reinspection of conforming group D sample units. Unless otherwise specified, sample units which have been subjected to and passed Group D inspection may be accepted on the contract or purchase order provided all visible and known damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.5.4.1.4 Voltage cycling. Voltage cycling shall be accomplished as follows: The input voltage shall be maintained at the nominal voltage during the first "on" cycle, at the high voltage (nominal + 10 percent) during the second cycle, and the low voltage (nominal - 10 percent) during the third "on" cycle. This sequence shall be repeated throughout the reliability test.

4.5.4.1.5 Vibration. For at least 30 minutes out of each 24 hours of operational "on" time the equipment shall be subjected to sinusoidal vibration at a nonresonant frequency between 20 and 60 Hz. The peak magnitude of the acceleration when measured on the equipment frame at the mounting point of the equipment shall be 2.2g ± 10%. The direction of vibration shall be that of the vibration critical axis (see 6.8).

4.5.4.2 Performance measurements. At least once daily the performance parameters shall be measured with the equipment stabilized at the high temperature. If the value of any parameter is not within tolerance, a failure shall be recorded and the required action shall be taken. The failure shall be presumed to have occurred immediately after the last successful measurement of the same parameter.

4.5.4.3 Accept reject criteria.

4.5.4.3.1 Test length. Testing will continue until the total unit hours together with the total count of relevant equipment failures permit either an accept or reject decision in accordance with the specified test plan. Only equipment "on" time may be used in MTBF determinations. Testing shall be monitored in such a manner that the times to failure may be estimated with reasonable accuracy. No single equipment "on" time shall be less than one half the average operating time of all equipments on test.

4.5.4.3.2 Pattern failures. Occurence of pattern failure (see 6.7.2) shall be cause for rejecting the lot.

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4.5.4.4 First article reliability test. The First Article reliability inspection testing shall be in accordance with Test Plan IIIC of MIL-STD-781.

4.5.4.5 Quality conformance (group D) reliability test. The reliability test for Group D inspection shall be in accordance with Test Plan XXIC of MIL-STD-781.

4.5.4.6 Reliability test conditions. The reliability test shall be in accordance with Table I, Ground Vehicle, of MIL-STD-781, with the exception that vibration shall be conducted in the non-operating transport mode and independent of the temperature environment. A failure is defined as the inability of the equipment to perform its required function. This includes catastrophic failures as well as any event which causes any departure from performance as required by this specification. Proper instrumentation shall be provided to insure immediate recognition of failure.

4.6 Burn-in. Each equipment submitted for acceptance shall be subjected to a 48 hour burn-in, as specified below. The last 24 hour cycle shall be failure free as evidenced by satisfactory completion of Group A electrical tests per Table I.

4.6.1 Burn-in conditions:

Chamber temperature: 100 ± 5°F

Line voltage: 1st 24 hours at 104 ± 0.5V, 2nd 24 hours at 126 ± 0.5.

Load: Short circuit on all channels, current meter set to 150 ± 2 mA on all channels.

Vibration: 22 Hz, 2.2g, non-operating, 30 minutes/day.

4.6.2 Burn-in procedure.

1. Mount equipment in chamber with axis of vibration perpendicular to base. Energize equipment, set timer to 19.9 hours and start time. Reset timer when time decreases below .1 and displays blanks. At some time during the 23.5 hour period turn equipment off and vibrate for a half hour, then turn equipment on. At the end of 24 hours measure and record current meter reading for each channel.

2. Repeat above procedure for second 24 hour period.

3. At end of the burn-in perform Group A electrical test.

4. In the event of a test failure take necessary corrective action and resubject unit to one 24 hour burn-in cycle.

4.7 Electrical performance characteristics. The Battery Charger, PP-7286/U shall be tested in accordance with the requirements of 3.7 and 3.4.4.

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4.8 Service test. (Environmental conditions) Unless otherwise specified the service conditions test shall be conducted in accordance with MIL-STD-810. Operational tests where required shall be the electrical tests specified for Group A inspection in Table II.

4.8.1 Altitude test. (see 3.8.1) The equipment shall be subjected to the test of Method 500, Procedure I of MIL-STD-810, except step 2 shall be 40,000 feet. The rate of change of altitude shall not exceed 2500 feet per minute.

4.8.2 Temperature tests. (see 3.8.2)

4.8.2.1 High temperature. The equipment shall be subjected to the test of Method 501, Procedure II, of Standard MIL-STD-810. For Step 4, raise internal temperature to 160°F (71°C). For Step 7, the internal chamber temperature shall be adjusted to 145°F (63°C) for the test item operation (this includes effects of solar radiation).

4.8.2.2 Low temperature. The equipment shall be subjected to the test of Method 502, Procedure I, of Standard MIL-STD-810. The storage temperature shall be -70°F (-57°C) and shall be maintained for a period of not less than 2 hours following stabilization of the test item. The low operating temperature shall be -25°F (-32°C).

4.8.3 Humidity test. (see 3.8.3) The equipment shall be subjected to the test of Method 507, Procedure II of MIL-STD-810.

4.8.4 Dust. (see 3.8.4) The equipment shall be subjected to the test of Method 510, Procedure I, of MIL-STD-810. The equipment shall be operated following the test.

4.8.5 Fungus. The fungus test shall be conducted in accordance with Method 508.2 Procedure of MIL-STD-810. A visual inspection shall be performed at the conclusion of this test: failure criteria is defined in 4.8.5.1.

4.8.5.1 Failure criteria, fungus. Operation of the equipment is not required but examination at the conclusion of the exposure shall reveal no evidence of viable fungus on any surface.

4.8.6 Vibration and shock. (see 3.8.6).

4.8.6.1 Vibration.

a. Part 1.

The equipment in its transit case shall be subjected to Method 514, Procedure XI, Part 2 of MIL-STD-810. Operate the test item and recore preperformance electrical test measurements. Inspect the test item mechanically.

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b. Part II.

1. Mounting techniques. The equipment in its combat transportation mode (in its transit case) shall be attached to the vibrator exciter table by its normal mounting means or by means of a rigid fixture capable of transmitting the vibration conditions as specified herein. Precautions shall be taken in the establishment of mechanical interfaces to minimize the introduction of undesirable responses in the test setup. Whenever possible the test load shall be distributed uniformly on the vibrator exiter table in order to reduce to the smallest amount, effects of unbalanced loads. Vibration amplitudes and frequencies shall be measured by techniques that will significantly affect test item input control or response. The input control sensing device(s) shall be rigidly attached to the vibration table or to the intermediate structure, if used, at or near as possible to the attachment point(s) of the test item.

2. Cycling. The test item shall be vibrated along each of its three mutually perpendicular axes in accordance with Test Level, Frequency Range, and Time Schedule outlined below. The frequency of the applied vibration shall be swept logarithmically over the specified frequency range.

Test level: 1.5 g's

Frequency range: 5 to 200 Hz

Time schedule: 84 minutes per axis

Sweep rate: 5 - 200 5 Hz in twelve (12) minutes

At the conclusion of the above test, visually inspect the test item for any evidence of mechanical damage and operate the test item and compare the results with preperformance test data.

4.8.6.2 Shock. The equipment shall be subjected to the tests of Method 516, Procedure II and V of MIL-STD-810.

4.8.7 Salt fog. (see 3.8.7) The equipment shall be subjected to the test of Method 509, Procedure I, of MIL-STD-810. Operation of the equipment is not required. Failure criteria shall be limited to corrosion, deterioration or change in tolerance limits of any internal or external parts which could in any manner prevent the test item from meeting operational service or maintenance requirements.

4.8.8 Immersion. (see 3.8.8) The equipment as prepared for field transport in its transit case shall be subjected to the test of Method 512, Procedure 1, of MIL-STD-810. This test shall be performed prior to and following vibration and shock testing of the same test item.

4.9 Electromagnetic interference test. (see 3.9) Test setups and test procedures shall conform to the measurement techniques of MIL-STD-462, Notice 3 as implemented by contractor supplied government approved test plan.

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4.9.1 Bonding and grounding. (see 3.9.1) Prior to the performance of EMI testing, compliance with the requirements of subparagraph 3.9.1 shall be ascertained. The bonding resistance measurements shall be recorded and included in the EMI test report.

4.10 Visual and mechanical inspection. (see 3.2, 3.3, 3.4.1, 3.13). Equipment shall be examined for the defects listed in MIL-STD-252.

4.11 Inspection for interchangeability. The dimensions listed below shall be gauged or measured to determine compliance with the physical interchangeability requirements of 3.12. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of cases, covers, and insertable subassemblies and replaceable parts, when such dimensions affect mating of parts.

(b) Internal dimensions of cavities in cases and assemblies, when such dimensions affect acceptability of insertable subassemblies.

(c) Location of fasteners on separate parts or assemblies which must mate together such as cases, covers and mountings.

4.12 Quality conformance inspection of packaging. Packaging shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of section 5. The rough handling test will be performed only when specified. (see 6.2c)

5. PACKAGING

5.1 Preservation. Preservation shall be as specified in MIL-STD-726, coded as follows:

5.1.1 Level A. JR-1-1-ØØ-00-NS-X-ED-Ø-00-A

5.1.2 Level B. JR-1-1-ØØ-00-NS-X-ED-Ø-00-B

5.2 Packing and marking. Packing and marking shall be in accordance with MIL-E-55585.

6. NOTES

6.1 Intended use. Battery Charger PP-7286/U is intended to charge a variety of small secondary batteries, including Nickel Cadmium battery BB-516/U as used in the AN/GVS-5 Laser Rangefinder, and other batteries used in Electronics Command equipment when powered with commercial or field generator 115/230V, 50/60/400 Hz Ac power.

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6.2 Ordering data. Procurement documents should specify:

a. Title, number and date of this specification and any amendment thereto.

b. Level A, Level B or Commercial Preservation and packing (see Section 5).

c. When rough handling test is required.

d. Marking and shipping requirements.

e. First article inspection.

(1) Number of units.
(2) First article test plan and test report(s) required.
(3) Action required for Group C Inspection failure.

f. Place of final inspection.

g. General instructions for conduct of reliability test or document indicating applicable portions of MIL-STD-781.

6.3 Color. The final paint film of Type 1 surface shall be Forest Green enamel, film designator W, per MIL-F-14072.

6.4 Group C and group D inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (see 4.5.3.2 and 4.5.4.3).

6.5 Nonmenclature. The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example PP-7286W/U. The contractor should apply for nomenclature in accordance with applicable clause in the contract. (see 1.1).

6.6 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.7 Failure definitions:

6.7.1 Equipment failure. The inability of the PP-7286/U Battery Charger to perform its function as specified herein to include damage resulting from inproper operation. Damage which results from abuse is not a failure nor is the need for adjustment of the operator's controls. The need for adjustment of controls not normally available to the operator is considered to be a failure. The performance requirements are not restricted to those of Table II.

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6.7.2 Pattern failure. The occurrence of two (2) or more failures of the same part in identical or equivalent application whose combined failure rate exceeds that predicted. When two (2) or more dependent failures of the same part occur whose combined failure rate exceeds that predicted and which are due to more than one independent failure, the dependent failures are relevant pattern failures. Unless a government approved Reliability Math Model is available failure rated found in MIL-HDBK-217 shall be used. (see 4.5.4.3.2)

6.7.3 Independent failure. A failure which will independently cause equipment performance outside of specified limits - one which occurs without being related to the failure of associated items.

6.7.4 Dependent failure. A failure of a part which is direct result of an independent failure - one which is caused by the failure of an associated item(s). Dependent failures are not necessarily present when simultaneous failures occur.

6.8 Vibration critical axis. The axis perpendicular to the base of the unit when the unit is in its normal operating position is the vibration critical axis. (see 4.5.4.1.5)

6.9 International standardization. Certain provisions of this specification are the subject of international standardization agreement (QSTAG 684). When amendment, revision or cancellation of this specification is proposed which will effect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels, including departmental standardization offices, if required.

Custodian:
Army - ER

Preparing activity:
Army - ER

(Project 6130-A212)

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-C-49144(ER) Charger, Battery PP-7286/U

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

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